Suggested problems: 1-15 odd

Problem 1: Find dz/dt.

(a) $z = x^2y - y^2$, $x = \sin(t)$, $y = e^t$

$$\frac{dz}{dt} = 2e^t \sin(t)\cos(t) + (\sin^2(t) - 2e^t)e^t$$

(b) $z = xy^2$, $x = e^{-t}$, $y = \sin(t)$

$$\frac{dz}{dt} = -\sin^2(t)e^{-t} + 2e^{-t}\sin(t)\cos(t)$$

(c) $z = x\sin(y) + y\sin(x)$, $x = t^2$, $y = \ln(t)$

$$\frac{dz}{dt} = (\sin(\ln(t)) + \ln(t)\cos(t^2))2t + (t^2\cos(\ln(t)) + \sin(t^2)) \cdot \frac{1}{t}$$

Problem 2: Let $z = f(x, y)$ where $x = g(t)$ and $y = h(t)$. Given the information below, find $dz/dt|_{t=1}$.

$f(3, 10) = 7$	$f(4, 11) = -20$
$f_x(3, 10) = 100$	$f_y(3, 10) = 0.1$
$f_x(4, 11) = 200$	$f_y(4, 11) = 0.2$
$f(3, 4) = -10$	$f(10, 11) = -1$
$g(1) = 3$	$h(1) = 10$
$g'(1) = 4$	$h'(1) = 11$

$$\left.\frac{dz}{dt}\right|_{t=1} = 100 \cdot 4 + \frac{1}{10} \cdot 11.$$

Problem 3: A steel bar with square cross sections 5 cm by 5 cm and length 3 meters is being heated. For each dimension, the bar expands 13×10^{-6} meters for each 1°C rise in temperature. What is the rate of change in the volume of the steel bar?

Let x denote the side length of a square cross section, and y the length of the beam. Then the volume is

$$V(x, y) = x^2y.$$

Thus

$$\frac{dV}{dt} = \frac{\partial V}{\partial x}\frac{dx}{dt} + \frac{\partial V}{\partial y}\frac{dy}{dt} = 2xy\frac{dx}{dt} + x^2\frac{dy}{dt}.$$

We are told that $dx/dt = dy/dt = 13 \times 10^{-6}$ when $x = 0.05$ and $y = 3$. Thus we have

$$\frac{dV}{dt} = 2(0.05)(3)(13 \times 10^{-6}) + (0.05)^2(13 \times 10^{-6}) = 3.9325 \times 10^{-6},$$

which has units equal to cubic meters per 1°C.

Problem 4: Find $\partial z/\partial u$ and $\partial z/\partial v$.

(a) $z = 2xy$, $x = u^2 + v^2$, $y = u/v$

$$\frac{\partial z}{\partial u} = 2 \cdot \frac{u}{v} \cdot 2u + 2(u^2 + v^2) \cdot \frac{1}{v}$$
$$\frac{\partial z}{\partial v} = 2 \cdot \frac{u}{v} \cdot 2v + 2(u^2 + v^2)\left(-\frac{u}{v^2}\right)$$

(b) $z = x^2 e^y$, $x = 4u$, $y = 3u^2 - 2v$

$$\frac{\partial z}{\partial u} = 2 \cdot 4ue^{3u^2-2v} \cdot 4 + 16u^2 e^{3u^2-2v} \cdot 6u$$
$$\frac{\partial z}{\partial v} = 16u^2 e^{3u^2-2v}(-2)$$